

Function analysis and graphs



Outline

Graph of real functions of a real variable

Examples

Graph of real functions of a real variable



Introduction: Function analysis

- ▶ In this section we put together limits, first derivatives and second derivatives to sketch the graph of a function
- ▶ We see the steps of function analysis and the useful additional information
- ▶ It is the synthesis of all the differential calculus seen so far



- ▶ The infinitesimal and differential calculus developed so far allows us to tackle completely the problem of drawing the graph of a function f , that is, to carry out **curve sketching** (function analysis).

The steps of function analysis (1/2)

1. Determine the (maximal) **domain** of f .
2. Compute the **limits at the boundary** of the domain.
Determine any **horizontal/vertical asymptotes** and **points of discontinuity**. Study the **sign of the function** and the **points where it vanishes** (possible only for relatively simple functions).
3. If the function tends to ∞ as $x \rightarrow \infty$, determine any **oblique asymptotes**.

The steps of function analysis (2/2)

4. Compute the **derivative function** f' at the points where it exists. Study the points where f is continuous but not differentiable and determine their nature (**corner points**, **inflection points with vertical tangent** or **cusps**). At corner points or at the endpoints of the domain it is useful to compute the **right or left derivatives**, which determine the **slope of the graph at those points**.
5. Study the sign of f' (on its domain) to obtain information on the **monotonicity** of f and on its **relative maximum and minimum points**. Then determine the **absolute maximum and minimum points**.

Additional information

1. It can be useful to determine an **asymptotic estimate at infinity** that tells us whether the function tends to ∞ in a *superlinear* or *sublinear* way or in a *linear* way (in which case it may have an oblique asymptote). The asymptotic estimate at infinity normally also gives information on the **concavity** of f at infinity.
2. Compute the **second derivative function** f'' and study its sign, to deduce information on the **concavity** and the **inflection points** of f .
3. Determine any symmetry of f (**even or odd functions**) and restrict the study to $x \geq 0$.
4. Determine any periodicity of f (**periodic functions**) and restrict the study to one period.

Summary

- ▶ **Steps:** domain, limits at the boundary and asymptotes, sign, oblique asymptotes, first derivative and points of non-differentiability, monotonicity and extrema
- ▶ **Additional information:** asymptotic estimates, second derivative (concavity and inflection points), symmetries, periodicity

Examples

Introduction: Examples

- ▶ In this section we apply the function analysis scheme step by step to two complete examples
- ▶ We will meet corner points, points with vertical tangent, horizontal, vertical and oblique asymptotes



Example (1/8)

Let us study the function and draw its graph:

$$f(x) = e^{-|x|} \sqrt{x^2 - 5x + 6}$$

We expect a corner point at $x = 0$ due to the presence of $|x|$, and points with vertical tangent where the radicand vanishes.

Example (2/8)

Step 1. The (maximal) domain of f is:

$$x^2 - 5x + 6 \geq 0, \quad (x - 3)(x - 2) \geq 0$$

$$\text{that is } x \in (-\infty, 2] \cup [3, +\infty)$$

Step 2. The limits at the boundary are:

$$\lim_{x \rightarrow +\infty} e^{-|x|} \sqrt{x^2 - 5x + 6} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 - 5x + 6}}{e^x} = 0^+$$

$$\lim_{x \rightarrow -\infty} e^{-|x|} \sqrt{x^2 - 5x + 6} = \lim_{x \rightarrow -\infty} \sqrt{e^{2x}(x^2 - 5x + 6)} = \lim_{y \rightarrow +\infty} \sqrt{\frac{y^2 + 5y + 6}{e^{2y}}} = 0^+$$

$$\lim_{x \rightarrow 3^+} e^{-|x|} \sqrt{x^2 - 5x + 6} = \lim_{x \rightarrow 3^+} e^{-x} \sqrt{(x - 3)(x - 2)} = 0^+$$

$$\lim_{x \rightarrow 2^-} e^{-|x|} \sqrt{x^2 - 5x + 6} = \lim_{x \rightarrow 2^-} e^{-x} \sqrt{(x - 3)(x - 2)} = 0^+$$

Example (3/8)

Hence the line $y = 0$ is a horizontal asymptote as $x \rightarrow \pm\infty$ and there are no vertical asymptotes. In the domain there are no points of discontinuity. We have $f(x) \geq 0$ for every x in the domain and $f(x) = 0$ for $x = 2$ and $x = 3$.

Step 3. The function has no oblique asymptotes.

Example (4/8)

Step 4. We compute the derivative function for $x \neq 0$:

$$\begin{aligned} f'(x) &= e^{-|x|} \left(-\operatorname{sgn}(x) \sqrt{x^2 - 5x + 6} + \frac{2x - 5}{2\sqrt{x^2 - 5x + 6}} \right) \\ &= \frac{e^{-|x|}}{2\sqrt{(x-3)(x-2)}} \left(-2\operatorname{sgn}(x)(x^2 - 5x + 6) + 2x - 5 \right) \end{aligned}$$

Or, equivalently:

$$f'(x) = \begin{cases} e^{-x} \cdot \frac{-2x^2 + 12x - 17}{2\sqrt{(x-3)(x-2)}} & \text{if } x > 0 \\ e^x \cdot \frac{2x^2 - 8x + 7}{2\sqrt{(x-3)(x-2)}} & \text{if } x < 0 \end{cases}$$

Example (5/8)

We compute the right and left derivatives at $x = 0$:

$$f'_+(0) = \lim_{x \rightarrow 0^+} \frac{e^{-x} (-2x^2 + 12x - 17)}{2\sqrt{(x-3)(x-2)}} = -\frac{17}{2\sqrt{6}}$$

$$f'_-(0) = \lim_{x \rightarrow 0^-} \frac{e^x (2x^2 - 8x + 7)}{2\sqrt{(x-3)(x-2)}} = \frac{7}{2\sqrt{6}}$$

hence $x = 0$ is a corner point, that is, $f'(0)$ does not exist.

Example (6/8)

We compute the right-hand limit at $x = 3$ and the left-hand limit at $x = 2$ of the derivative function:

$$f'_+(3) = \lim_{x \rightarrow 3^+} \frac{e^{-x} \overbrace{(-2x^2 + 12x - 17)}^{\rightarrow 1}}{2\sqrt{(x-3)(x-2)}} = +\infty$$

$$f'_-(2) = \lim_{x \rightarrow 2^-} \frac{e^{-x} \overbrace{(-2x^2 + 12x - 17)}^{\rightarrow -1}}{2\sqrt{(x-3)(x-2)}} = -\infty$$

Hence the function has points with vertical tangent at $x = 2$ and $x = 3$.

Example (7/8)

Step 5. We study the sign of the derivative function:

► For $x > 0$ we have:

$$f'(x) \geq 0 \quad \text{if} \quad 2x^2 - 12x + 17 \leq 0,$$

$$\text{that is } x \in \left[\underbrace{\frac{6 - \sqrt{2}}{2}}_{\approx 2.3}, \underbrace{\frac{6 + \sqrt{2}}{2}}_{\approx 3.7} \right]$$

hence:

$$f \text{ is decreasing for } x \in [0, 2] \cup \left[\frac{6 + \sqrt{2}}{2}, +\infty \right), \quad f$$

$$\text{is increasing for } x \in \left[3, \frac{6 + \sqrt{2}}{2} \right]$$

Example (8/8)

We have $f' \left(\frac{6+\sqrt{2}}{2} \right) = 0$, hence $x = \frac{6+\sqrt{2}}{2}$ is a relative maximum point. The relative maximum is $f \left(\frac{6+\sqrt{2}}{2} \right) \approx 0.027$

► For $x < 0$ we have:

$$f'(x) \geq 0 \quad \text{if} \quad 2x^2 - 8x + 7 \geq 0,$$

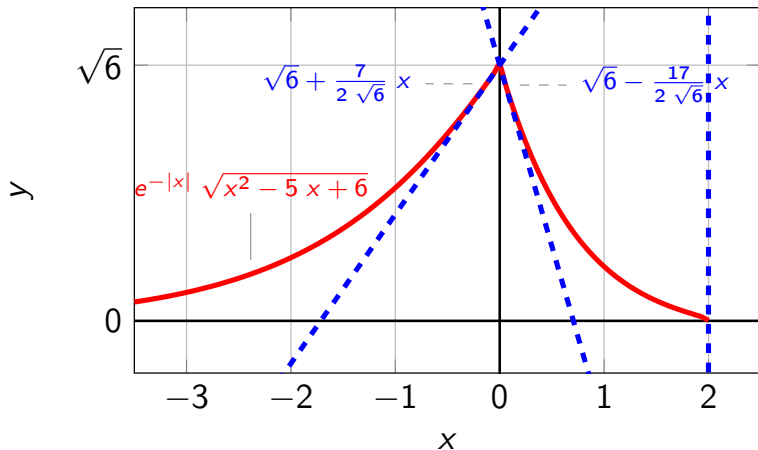
$$\text{that is, if } x \in \left[-\infty, \frac{4 - \sqrt{2}}{2} \right] \cup \left[\frac{4 + \sqrt{2}}{2}, +\infty \right]$$

hence $f(x)$ is increasing for $x < 0$.

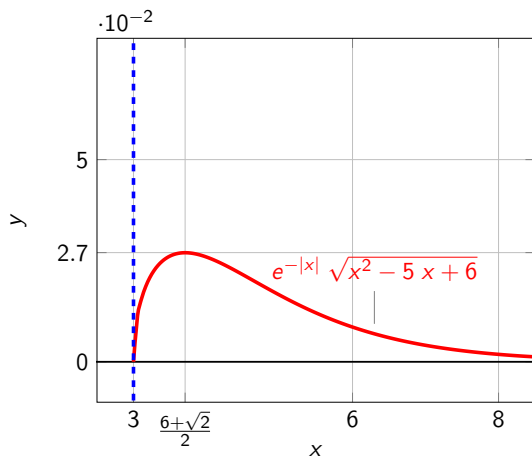
We can deduce that $x = 0$ (where f is not differentiable) is an absolute maximum point; the absolute maximum is $\sqrt{6} \approx 2.44$.

There are also two inflection points, one in $(0, 2)$ and one in $(3, +\infty)$, which can be obtained by studying the sign of the second derivative.

Example



Example



Example (1/10)

Let us study the function and draw its graph:

$$f(x) = x \cdot \exp\left(\frac{x+2}{x-1}\right)$$

Step 1. The (maximal) domain of f is:

$$x \in \mathbb{R} \setminus \{1\}$$

Example (2/10)

Step 2. The limits at the boundary are:

$$\lim_{x \rightarrow +\infty} x \cdot \exp\left(\frac{x+2}{x-1}\right) = +\infty$$

$$\lim_{x \rightarrow -\infty} x \cdot \exp\left(\frac{x+2}{x-1}\right) = -\infty$$

$$\frac{x+2}{x-1} \rightarrow +\infty$$

$$\text{as } x \rightarrow 1^+, \quad \text{hence } \lim_{x \rightarrow 1^+} x \exp\left(\frac{x+2}{x-1}\right) = +\infty$$

$$\frac{x+2}{x-1} \rightarrow -\infty \text{ as } x \rightarrow 1^-, \quad \text{hence } \lim_{x \rightarrow 1^-} x \exp\left(\frac{x+2}{x-1}\right) = 0$$

Hence there are no horizontal asymptotes and $x = 1$ is a point of discontinuity. Moreover, $x = 1$ is a vertical asymptote as $x \rightarrow 1^+$. We have $f(x) \geq 0$ for $x \geq 0$ and $f(x) = 0$ for $x = 0$.

Example (3/10)

Step 3. We compute an asymptotic estimate; we have:

$$\frac{x+2}{x-1} \rightarrow 1 \quad \text{as } x \rightarrow +\infty, \quad \frac{x+2}{x-1} \rightarrow 1 \quad \text{as } x \rightarrow -\infty$$

hence

$$x \cdot \exp\left(\frac{x+2}{x-1}\right) \sim x \cdot e$$

as $x \rightarrow \pm\infty$ therefore f has linear growth

We therefore check for the presence of oblique asymptotes. We try to compute the following limit:

$$\lim_{x \rightarrow \pm\infty} \left(x \cdot \exp\left(\frac{x+2}{x-1}\right) - x e \right) = \lim_{x \rightarrow \pm\infty} x \cdot e \cdot \left(\exp\left(\frac{x+2}{x-1} - 1\right) - 1 \right)$$

Example (4/10)

We have

$$\frac{x+2}{x-1} - 1 = \frac{3}{x-1} \rightarrow 0 \text{ as } x \rightarrow \pm\infty$$

therefore

$$\exp\left(\frac{x+2}{x-1} - 1\right) - 1 \sim \frac{3}{x-1}$$

$$\text{as } x \rightarrow \pm\infty; \quad x \cdot e \cdot \left(\exp\left(\frac{x+2}{x-1} - 1\right) - 1\right) \sim \frac{3 \cdot x \cdot e}{x-1} \text{ as } x \rightarrow \pm\infty$$

consequently:

$$\lim_{x \rightarrow \pm\infty} \left(x \cdot \exp\left(\frac{x+2}{x-1}\right) - x \cdot e \right) = \lim_{x \rightarrow \pm\infty} \frac{3 \cdot x \cdot e}{x-1} = 3 \cdot e$$

Hence the function has the oblique asymptote $y = e x + 3 e$ as $x \rightarrow \pm\infty$.

Example (5/10)

Step 4. We compute the derivative function for $x \neq 1$:

$$\begin{aligned} f'(x) &= \exp\left(\frac{x+2}{x-1}\right) \cdot \left(1 + x \cdot \frac{(x-1) - (x+2)}{(x-1)^2}\right) = \exp\left(\frac{x+2}{x-1}\right) \cdot \frac{(x-1)^2 - 3x}{(x-1)^2} \\ &= \exp\left(\frac{x+2}{x-1}\right) \cdot \frac{x^2 - 5x + 1}{(x-1)^2} \end{aligned}$$

For every $x \neq 1$, f' is defined. We compute the left-hand limit at $x = 1$:

$$f'_-(1) = \lim_{x \rightarrow 1^-} \exp\left(\frac{x+2}{x-1}\right) \cdot \frac{x^2 - 5x + 1}{(x-1)^2} = 0$$

the exponential goes to zero faster than $(x-1)^2$; hence the graph reaches $x = 1$ with a horizontal tangent, from the left.

Example (6/10)

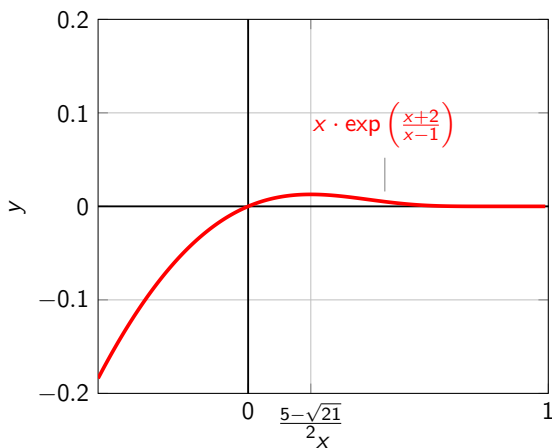
Step 5. We study the sign of the derivative function:

$$f'(x) \geq 0 \quad \text{for} \quad x^2 - 5x + 1 \geq 0$$

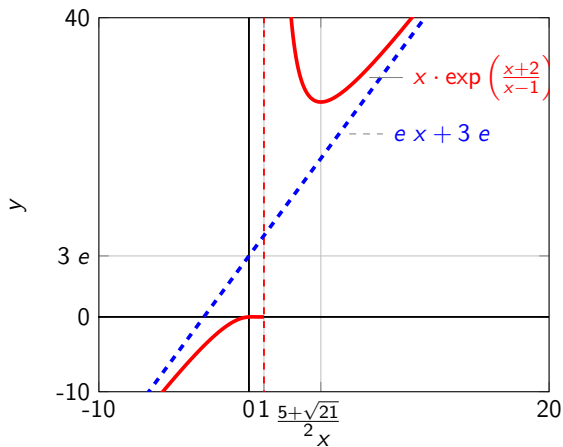
$$\text{that is } x \in \left[-\infty, \frac{5 - \sqrt{21}}{2} \right] \cup \left[\frac{5 + \sqrt{21}}{2}, +\infty \right]$$

hence $x = \frac{5 + \sqrt{21}}{2}$ is a relative minimum point and $x = \frac{5 - \sqrt{21}}{2}$ is a relative maximum point. There is an inflection point in $\left(\frac{5 - \sqrt{21}}{2}, 1 \right)$, which can be obtained by studying the sign of the second derivative.

Example (7/10)



Example (8/10)



Example (9/10)

We have $f'_-(1) = 0$ since:

$$f'_-(1) = \lim_{x \rightarrow 1^-} \exp\left(\frac{x+2}{x-1}\right) \cdot \frac{x^2 - 5x + 1}{(x-1)^2} = \lim_{x \rightarrow 1^-} \frac{\exp\left(\frac{x+2}{x-1}\right)}{(x-1)^2} \cdot \underbrace{(x^2 - 5x + 1)}_{\rightarrow -3 \text{ as } x \rightarrow 1^-}$$

with the change of variable $y = \frac{x+2}{x-1}$: if $x \rightarrow 1^-$ then $y \rightarrow -\infty$;
moreover:

$$y = \frac{x+2}{x-1}, \quad y = \frac{x-1+3}{x-1}, \quad y = 1 + \frac{3}{x-1}$$

$$\text{and } x = 1 + \frac{3}{y-1}$$

Example (10/10)

hence we have:

$$\lim_{x \rightarrow 1^-} \frac{\exp\left(\frac{x+2}{x-1}\right)}{(x-1)^2} = \lim_{y \rightarrow -\infty} \frac{e^y}{\left(\frac{3}{y-1}\right)^2} = \frac{1}{9} \lim_{y \rightarrow -\infty} e^y \cdot \underbrace{(y-1)^2}_{\sim y^2 \text{ as } y \rightarrow -\infty}$$

with a second change of variable $z = -y$: if $y \rightarrow -\infty$ then $z \rightarrow +\infty$ and we have

$$\frac{1}{9} \lim_{y \rightarrow -\infty} e^y \cdot y^2 = \frac{1}{9} \lim_{z \rightarrow +\infty} e^{-z} \cdot (-z)^2 = \frac{1}{9} \lim_{z \rightarrow +\infty} \frac{z^2}{e^z} = 0$$

by the theorem on the hierarchy of infinities.

Summary

- ▶ $f(x) = e^{-|x|}\sqrt{x^2 - 5x + 6}$: corner point at $x = 0$ (absolute maximum), vertical tangents at $x = 2$ and $x = 3$, horizontal asymptote $y = 0$
- ▶ $f(x) = x \exp\left(\frac{x+2}{x-1}\right)$: vertical asymptote $x = 1$ (from the right), oblique asymptote $y = ex + 3e$, one relative maximum and one relative minimum

Final summary

- ▶ Function analysis combines domain, limits, asymptotes, first and second derivative
- ▶ Right and left derivatives clarify the behaviour at points of non-differentiability
- ▶ Asymptotic estimates and changes of variable simplify the computation of the required limits